

AVI GUPTA

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Summary

I am a fifth-year PhD student at IIIT Delhi, advised by Dr. Koteswar Rao Jerripothula and Dr. Tammam Tillo. My research interests include Computer Vision and Deep Learning. In particular, I work in Semantic Understanding of Large Multimodal Models, Continual Learning, and Camouflaged Scene Understanding.

Education

Indraprastha Institute of Information Technology, Delhi <i>Doctoral Student (Ph.D.) in Computer Science</i>	Aug. 2022 – Present <i>Delhi, Delhi</i>
Dr. Akhilesh Das Gupta Institute of Technology & Management, Delhi <i>Bachelor of Technology in Information Technology (8.56/10)</i>	Aug. 2017 – Jul. 2021 <i>Delhi, Delhi</i>

Relevant Coursework

- Machine Learning
- Advanced Machine Learning
- Advances in Deep Learning
- Artificial Intelligence
- Computer Vision
- Digital Image Processing

Teaching Experience

- Meta-Learning
- Database Management System
- Introduction to Programming
- Digital Image Processing
- Data Structures and Algorithms
- Artificial Intelligence

Industry Experience

Dolby Laboratories <i>Research Intern</i>	Jan 2026 – Apr 2026 <i>Bangalore, Karnataka</i>
Worked on efficient Class-Incremental Learning in Audio-Visual Multimodality to deploy on edge devices.	
GE Aerospace <i>Research Intern</i>	Aug 2025 – Dec 2025 <i>Bangalore, Karnataka</i>
1) Built an LLM-powered pipeline (via AWS Bedrock) to extract and structure unstructured aircraft engine data (via AWS S3), storing results in a SQL database for downstream processing.	
2) Leveraged X-ray images of engine blades to detect and localize the patterns of scratches, dents, etc.	
Tata Consultancy Services <i>Cloud Developer (SME)</i>	Jul 2021 – Aug 2022 <i>Delhi, Delhi</i>
NFV OpenStack technology to deploy cloud services (Python, OpenStack, Shell Scripting, OpenSource Mano)	
Digital Green <i>Backend Developer Intern</i>	Jun 2019 – Aug 2019 <i>Delhi, Delhi</i>
Script Development for Automation of the Monitoring mailer for Data and Revenue reconciliation.	

Research Projects

Instance-based Similarity using Large Vision Model <i>Ongoing</i>	Sept 2026 – Present
Working on retrieving and segmenting images through instance-based similarity using a large vision model like SAM.	
Toward Genuine Forgetting: Selective Unlearning in Large Language Models <i>Ongoing</i>	Aug 2026 – Present
Working on understanding how to unlearn requested knowledge in LLMs in a continual setting.	
Referring Camouflaged Object Discovery <i>Extension Ongoing for Journal</i>	Sept 2025 – Present <i>WACV 2025</i>
Working on extending the published work to incorporate videos and other scenarios.	

Publications

Avi Gupta, Saurabh Yadav, Koteswar Rao Jerripothula, and Tammam Tillo, "SELECT: SELEctive Context Transfer in Class-Incremental Semantic Segmentation," in *Proceedings of British Machine Vision Conference (BMVC)*, 2026. 🌐

Avi Gupta, Nilotpal Sinha, Vishnu Raj, Sambuddha Saha, Pratik Joshi, Koteswar Rao Jerripothula, and Tammam Tillo. "Listen, Look, and Learn: Learning Without Forgetting through SAM-Audio", in *International Conference on Machine Learning Workshops (ICMLw)*, 2026. 🌐

Saurabh Yadav¹, **Avi Gupta**¹ and Koteswar Rao Jerripothula, "SAMWave: Adapting Segment Anything Model to Difficult Tasks," in *Proceedings of British Machine Vision Conference (BMVC)*, 2025 **(Oral)** 🌐

Avi Gupta, Koteswar Rao Jerripothula and Tammam Tillo, "CIRCOD: Co-saliency Inspired Referring Camouflaged Object Discovery," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2025. 🌐

Shyama Sastha, **Avi Gupta**, and Pankaj Jalote. "An Experience Report on Teaching a Large Introductory Programming Course in Hybrid Mode." In *Proceedings of the 55th ACM Technical Symposium on Computer Science Education V. 2*, pp. 1704-1705. 2024. 🌐

Under-Review

Toward Genuine Forgetting: Selective Unlearning in Large Language Models **2026**
Understanding how to unlearn requested knowledge in LLMs in a continual setting.

Generalizing Multimodal Foundational models for Class-Incremental Learning **2026**
Generalizing different foundational multimodal models in a context-driven class-incremental setting.

Camouflaged Object Detection via Multispectral Imaging 🌐 **2026**
Introduce a novel approach that detects camouflaged objects from the corresponding multispectral inputs and predict corresponding binary mask.

SAMwave: Wavelet-Driven Feature Enrichment for Effective Adaptation of Segment Anything Model **IJCV**
Proposed real and complex-wavelet transformation for SAM, SAM2 and SAM3 on low-level vision tasks. **2026**
Saurabh Yadav¹, **Avi Gupta**¹ and Koteswar Rao Jerripothula

Reviewer

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|-----------|--------|---------|----------|---------|
| • NeurIPS | • ECCV | • TCSVT | • ICASSP | • CVPRW |
| • CVPR | • TMLR | • BMVC | • ICPR | • ICMLW |

Achievements

Doctoral Consortium **2026**
Will be presenting my thesis work at the Doctoral Consortium BMVC 2026.

DEI Grant **2026**
Received a full fee waiver to attend BMVC 2026 (14% acceptance).

Anveshan Setu Fellowship **2025**
Received a fellowship to work under a professor for a month.

Travel grant **2024**
Received a travel grant to present my work at BITS Goa IndoML 2024

GATE **2021, 2022**
Qualified

References

Dr Tammam Tillo (TYUST, China) ✉ 🌐	<i>PhD Supervisor</i>
Dr Koteswar Rao Jerripothula (IIT, Kanpur) ✉ 🌐	<i>PhD Supervisor</i>
Dr Gautam Shroff (IIIT, Delhi) ✉ 🌐	<i>Academic Mentor</i>

¹Equal Contribution